



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies

- Data Collection
- Data wrangling
- EDA with Data Visualization
- EDA with SQL
- Build an Interactive Map with Folium
- Build a Dashboard with Plotly Dash
- Predictive Analysis (Classification)

- Summary of all results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Introduction

- Project background and context

Space X advertises a Falcon 9 rocket launch on its website at a cost of \$62 million. Other providers charge up to \$165 million each, much of the savings being that Space X can reuse the first stage. Therefore, if we can determine whether the first stage will land, we can determine the cost of the launch. This information can be used to create a machine learning model to predict whether the first stage will land for reuse.

- Problems you want to find answers

1. What is the trend of successful Falcon 9 rocket launches from year to year?
2. How do Flight Number, launch site, payload mass, and orbit affect launch outcomes?
3. What is the best algorithm for creating a machine learning model to predict the success or failure of a first stage landing?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Using SpaceX API
 - Using Web Scraping
- Perform data wrangling
 - Dealing with Missing Values
 - Create a landing outcome label from Outcome column
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Building and tuning classification models using several different machine learning algorithms
 - Classification model evaluation using confusion matrix

Data Collection

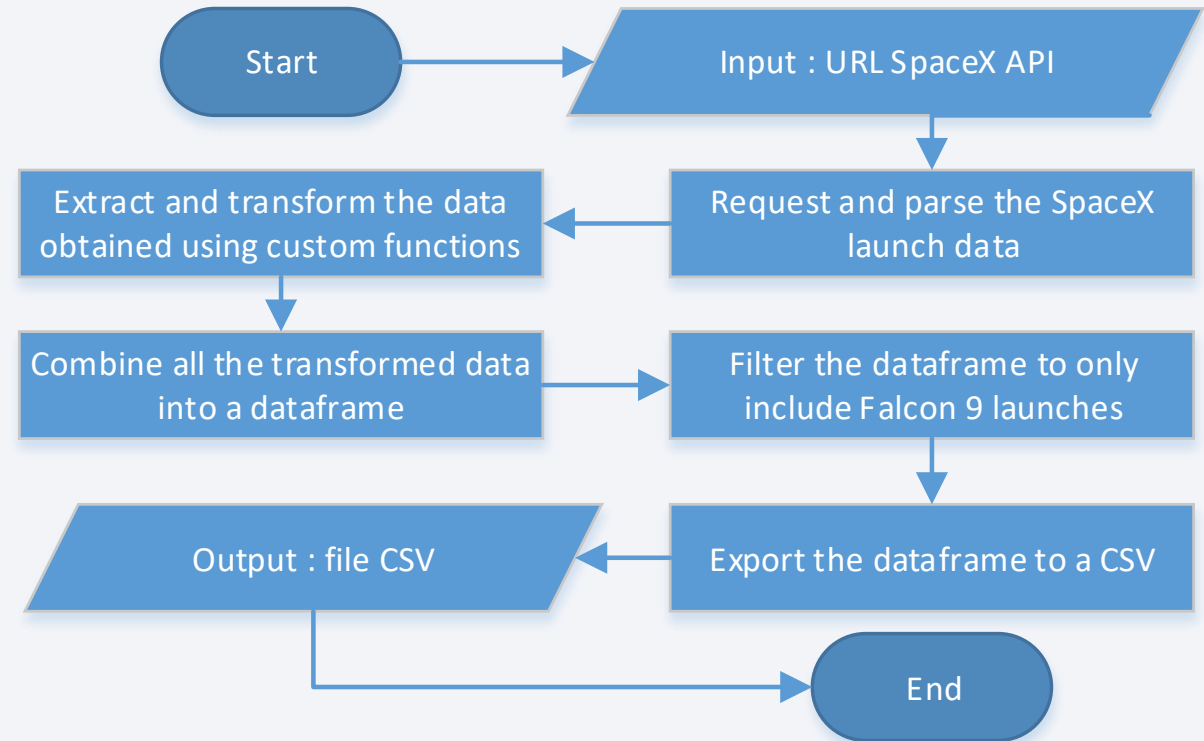
The data collected is data from the launch of the Falcon 9 rocket. Data retrieval uses 2 different methods, namely using the SpaceX API and Web Scraping.

Data collection using the SpaceX API is done by making a request to the SpaceX Rest API while data collection using Web Scraping is done by taking data from the Wikipedia website page.

Metode	SpaceX API	Web Scraping
Collected Data	FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, and Latitude	Flight No., Date, Time, Version Booster, Launch site, Payload, Payload mass, Orbit, Customer, Launch outcome, and Booster landing
FlowChart	<pre>graph TD; Start([Start]) --> Input[/Input : URL SpaceX API/]; Input --> Request[Request and parse the SpaceX launch data]; Request --> Transform[Extract and transform the data obtained using custom functions]; Transform --> Combine[Combine all the transformed data into a dataframe]; Combine --> Filter[Filter the dataframe to only include Falcon 9 launches]; Filter --> Export[Export the dataframe to a CSV]; Export --> Output[/Output : file CSV/]; Output --> End([End]);</pre>	<pre>graph TD; Start([Start]) --> Input[/Input : URL Wikipedia page/]; Input --> Request[Request the Falcon 9 launch data from Input]; Request --> Transform[Extract and transform data from the HTML table header]; Transform --> Combine[Combine all the data obtained into a dataframe]; Combine --> Export[Export the dataframe to a CSV]; Export --> Output[/Output : file CSV/]; Output --> End([End]);</pre>

Data Collection – SpaceX API

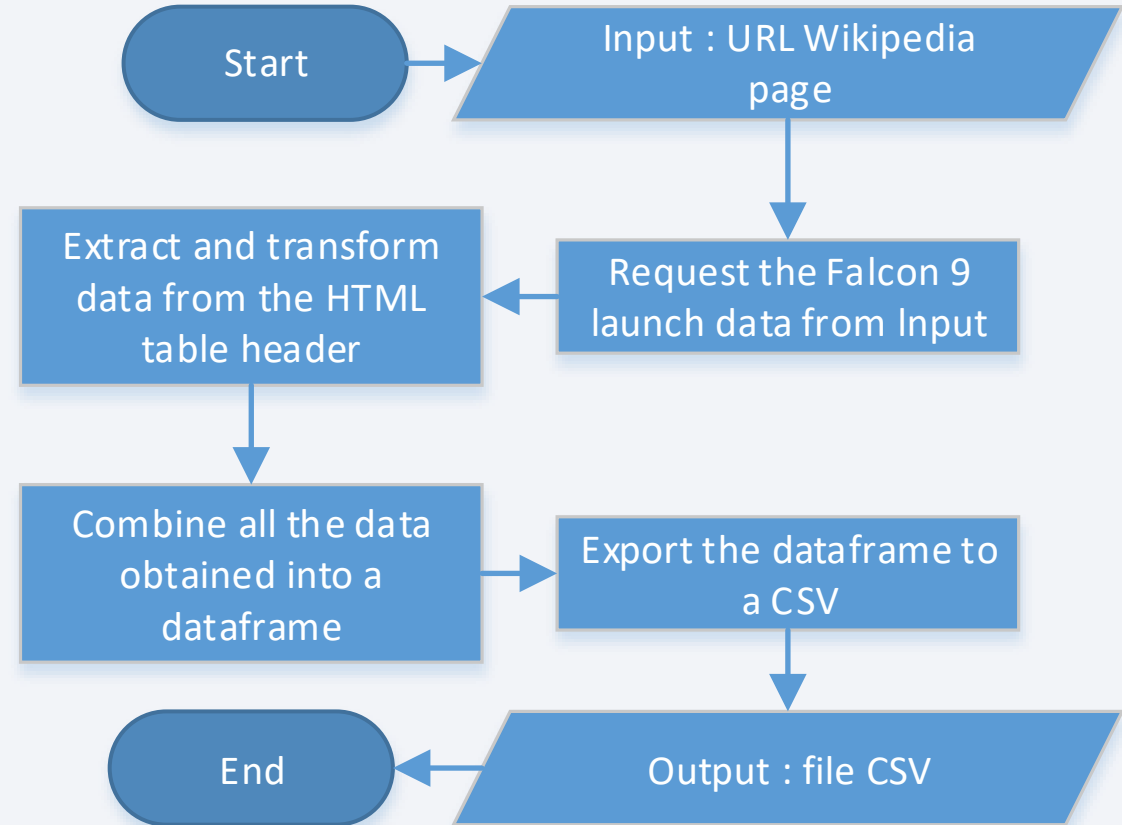
- Request SpaceX launch data using the SpaceX API via the URL "https://api.spacexdata.com/v4/launches/past".
- Extract the data obtained by encoding the data as JSON using `.json()` and turn it into a dataframe using `.json_normalize()`.
- Transforming the dataframe to make the data analysis process easier, because there is some data that only contains IDs such as rocket, payloads, launchpad, and cores, so it is necessary to request the data again using the SpaceX API via the URL "https://api.spacexdata.com/v4/rockets/id_rocket" with `id_rocket` using value from column `rocket` in dataframe.
- After all the data has been requested and extracted, then combine all the required data into one dataframe.
- Filter the dataframe to only include Falcon 9 launches
- Export the dataframe to file CSV.



[LINK GITHUB](#)

Data Collection - Scraping

- Request the Falcon 9 launch data from Wikipedia page using web scraping data with BeautifulSoup via the URL "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922".
- Extract the obtained data, find the data that has the table tag, and transform the data to make the data analysis process easier.
- After all the data has been requested and extracted, then combine all the required data into one dataframe.
- Export the dataframe to file CSV.

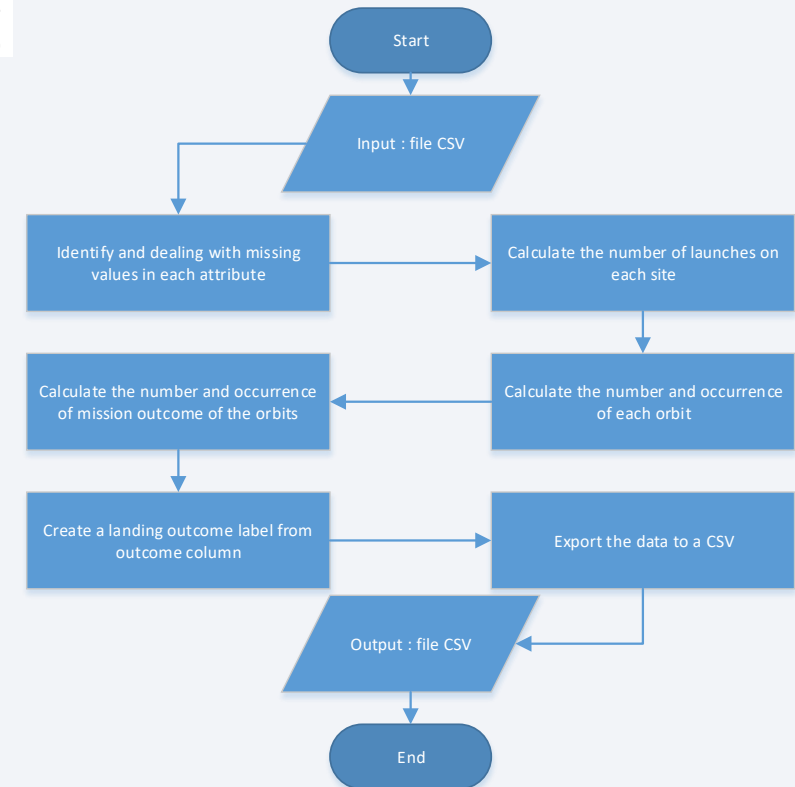


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Data Wrangling

- Load the csv file created from the previous step.
- Identify missing values for each existing attribute, in this case there are 2 attributes that have missing values, namely PayloadMass and LandingPad, for PayloadMass the missing value will be filled with the average value while for LandingPad the missing value will retain the none value to represent when the landing pad is not used.
- Calculate the number of launches at each site, the number and occurrence of each orbit, and the mission outcome from those orbits.
- Create a result label with the result attribute / column, in this case the label you want to create will represent the success or failure of the landing outcome, outcomes that represent success include True ASDS, True RTLS, and True Ocean while outcomes that represent failure include None None, False ASDS, False Ocean, None ASDS, and False RTLS.
- Export the data to a file CSV.

LaunchSite		
CCAFS	SLC 40	55
KSC	LC 39A	22
VAFB	SLC 4E	13
Orbit		
GTO		27
ISS		21
VLEO		14
PO		9
LEO		7
SSO		5
MEO		3
HEO		1
ES-L1		1
SO		1
GEO		1
Outcome		
True	ASDS	41
None	None	19
True	RTLS	14
False	ASDS	6
True	Ocean	5
False	Ocean	2
None	ASDS	2
False	RTLS	1



[LINK GITHUB](#)

EDA with Data Visualization

Chart	Data	Explanations
Scatter point	○ Flight number vs launch site ○ Payload mass vs launch site ○ Flight number vs orbit type ○ Payload mass vs orbit type	Shows the relationship between two variables by looking at the distribution of points on the diagram to find out the pattern and correlation of the data
Bar chart	Success rate of each orbit type	Shows the relationship between the values of each category of a variable to see the distribution of the data.
Line chart	Launch success yearly trend	Shows the relationship between variables and time series by looking at the lines on the diagram can find trends in the data.

[LINK GITHUB](#)

EDA with SQL

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first successful landing outcome in ground pad was achieved
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List the names of the booster versions which have carried the maximum payload mass
- List the records which will display the month names, failure landing outcomes in drone ship, booster versions, launch site for the months in year 2015
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Build an Interactive Map with Folium

Map	Objects	Explanations
Mark all launch sites on a map	○ Markers ○ Circles	Shows a map with markers on each launch site and a circle with a radius of 1000 meters with the launch site as the center point.
Mark the success or failed launches for each site on the map	○ Markers ○ Circles	Shows the same map as before but adds color to the markers which will show the class with green for successful launches and red for failed launches.
Calculate the distances between a launch site to its proximities	○ Markers ○ Circles ○ Lines	Shows the same map as before but adds lines showing the distance between the launch site and the coastline, railway, highway, and city.

[LINK GITHUB](#)

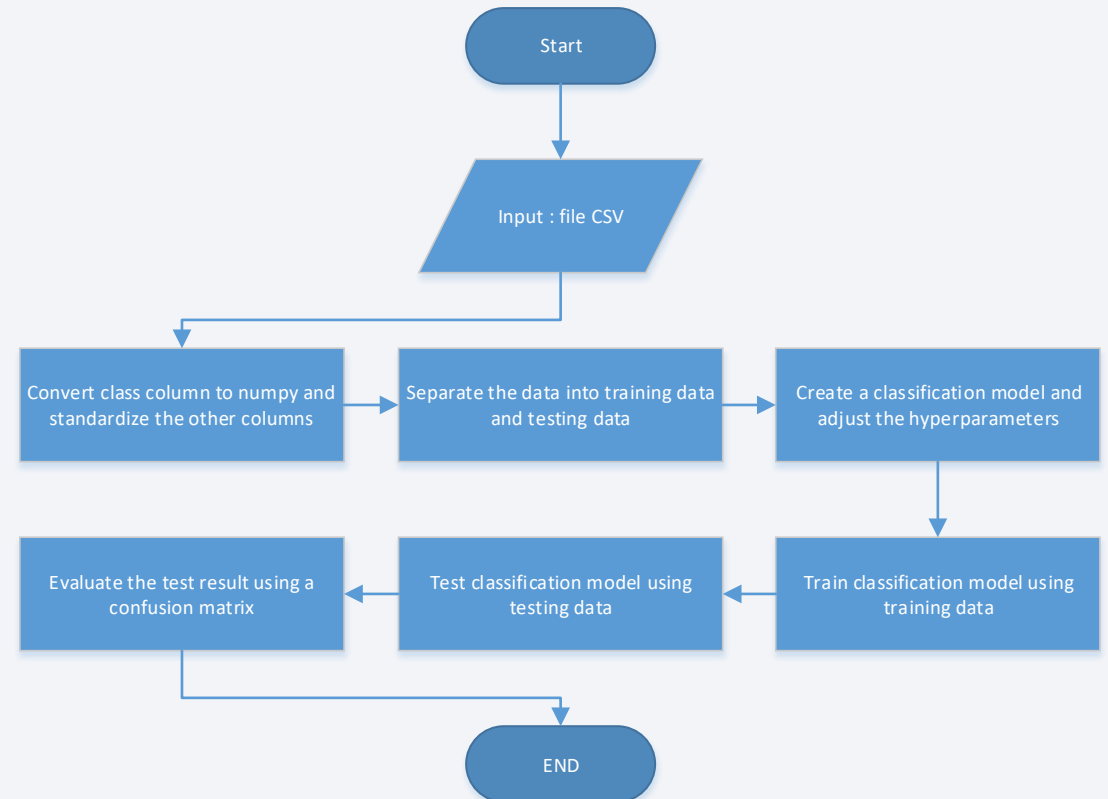
Build a Dashboard with Plotly Dash

Plot	Interactions	Explanations
Pie Chart	-	Displays the percentage of successful launches from each launch site.
	Dropdown List	Displays the percentage of successful and failed launches from the launch sites selected in the dropdown list (if you select all sites it will display a pie chart like without interactions)
Scatter Point	-	Shows the relationship between launch outcome and payload mass.
	Dropdown List	The data displayed will be filtered based on the launch site selected in the dropdown list.
	Slider	The data displayed will be filtered based on the payload mass with minimum and maximum values according to the slider.

[LINK GITHUB](#)

Predictive Analysis (Classification)

- Load the CSV file created from the previous step.
- Convert the class column to numpy using `to_numpy()` and standardize the other columns using `StandardScaler()`.
- Separate the data into training and testing data using `train_test_split()`, 80% training data and 20% testing data.
- Create a classification model using the logistic regression, SVM, KNN, and Tree algorithms, then adjust the hyperparameters using `GridSearchCV()`.
- Train the classification model using the training data, then test the classification model using the testing data.
- Evaluate the test results using a confusion matrix and compare the results of all the classification models created.



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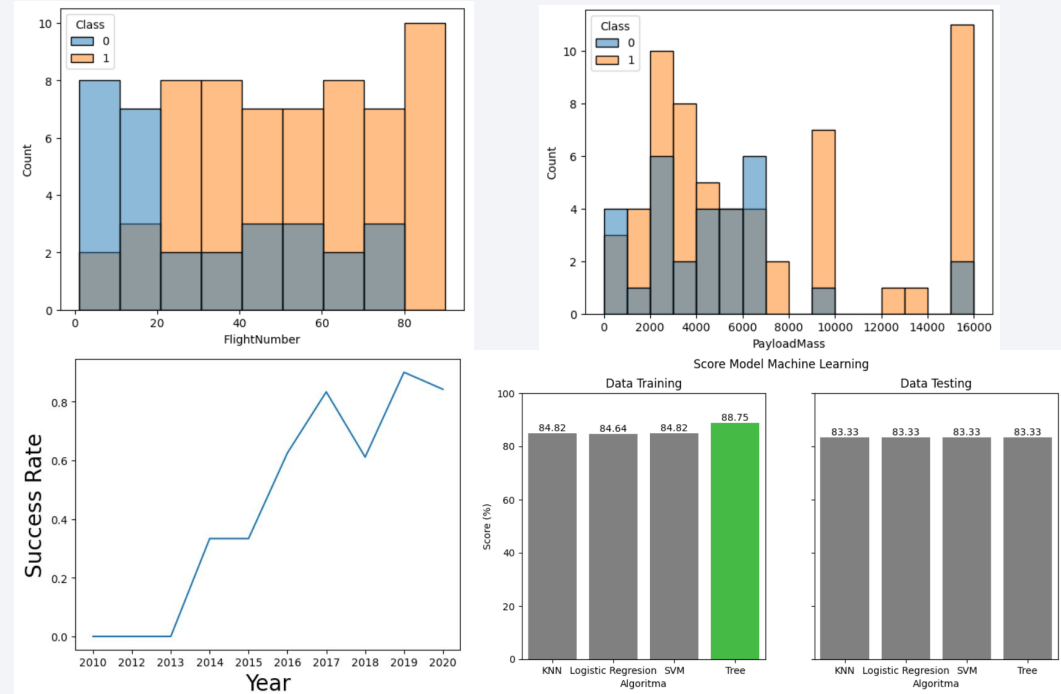
Results

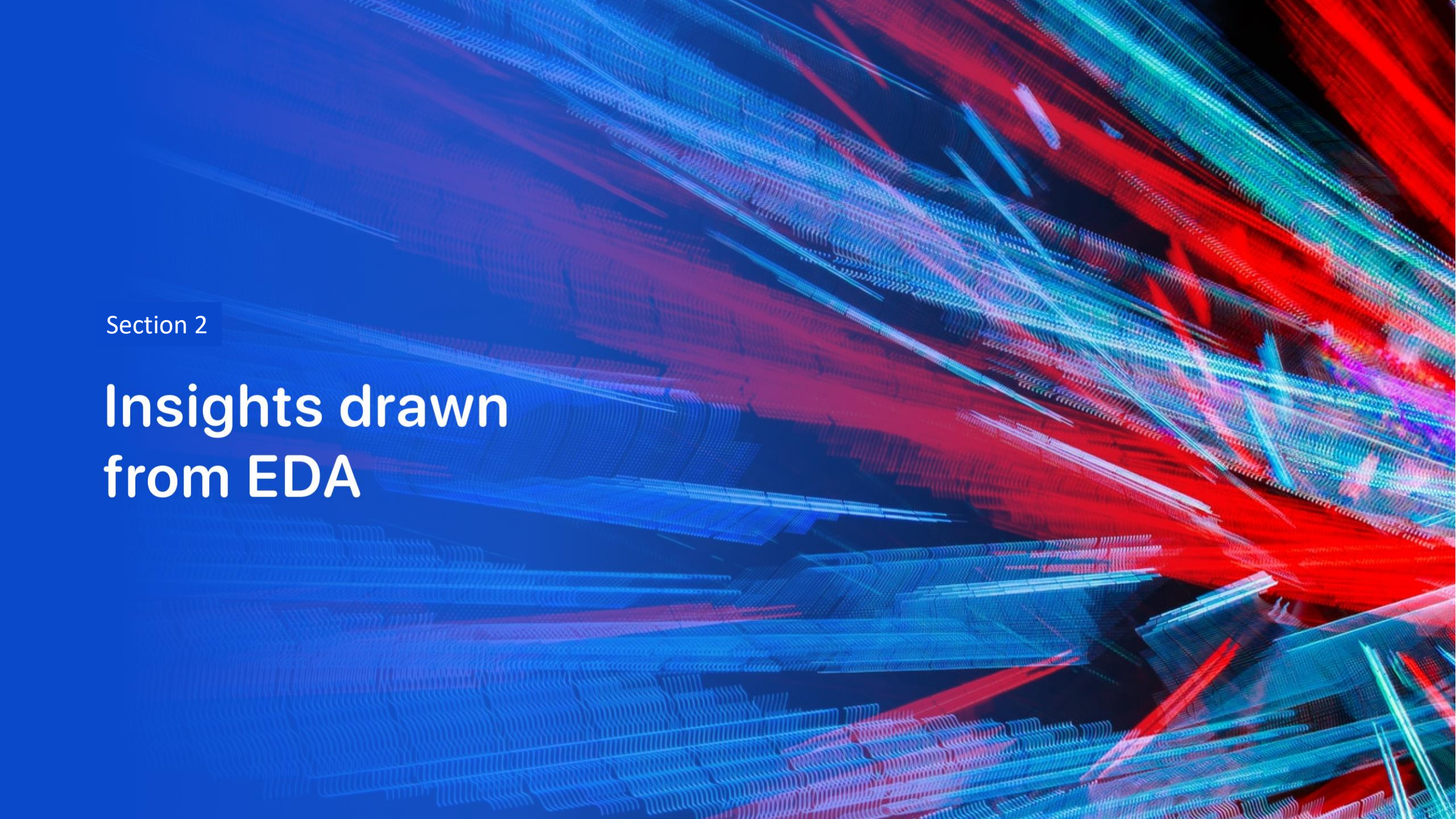
- Exploratory data analysis results
 - The higher the Flight Number or the greater the Payload Mass, the higher the success rate.
 - Every year, the success rate has been on an upward trend.

- Predictive analysis results

The best algorithm for prediction on this data is the Tree algorithm with an accuracy of 88.75% on training data and 83.33% on test data.

- Interactive analytics demo in screenshots

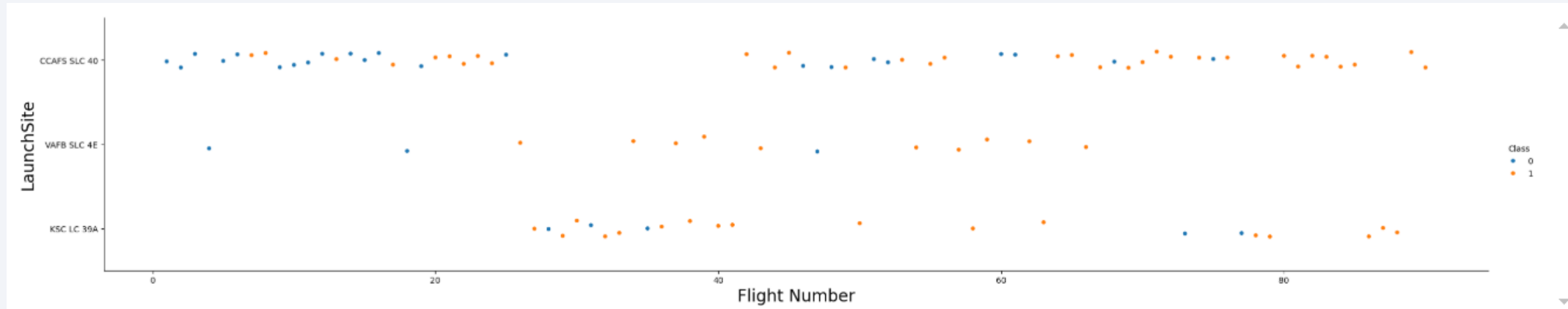


The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

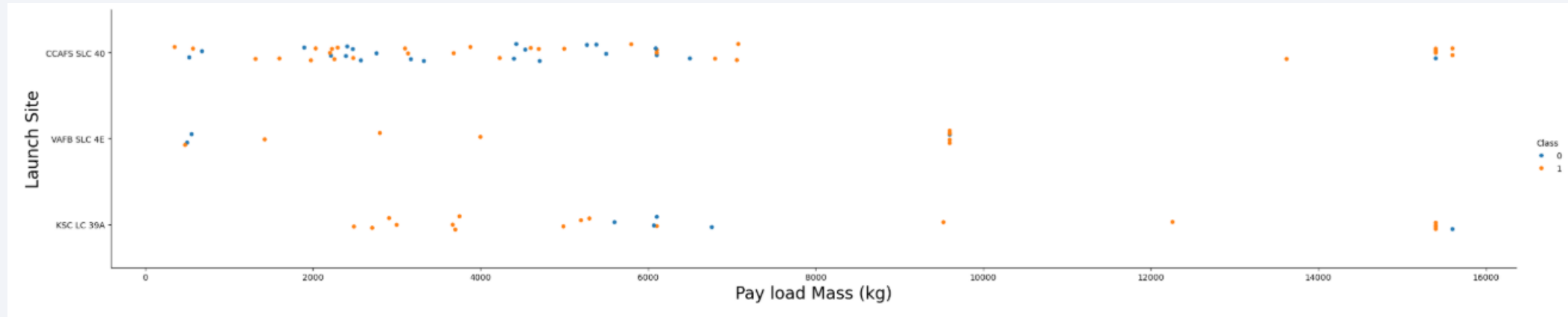
Insights drawn from EDA

Flight Number vs. Launch Site



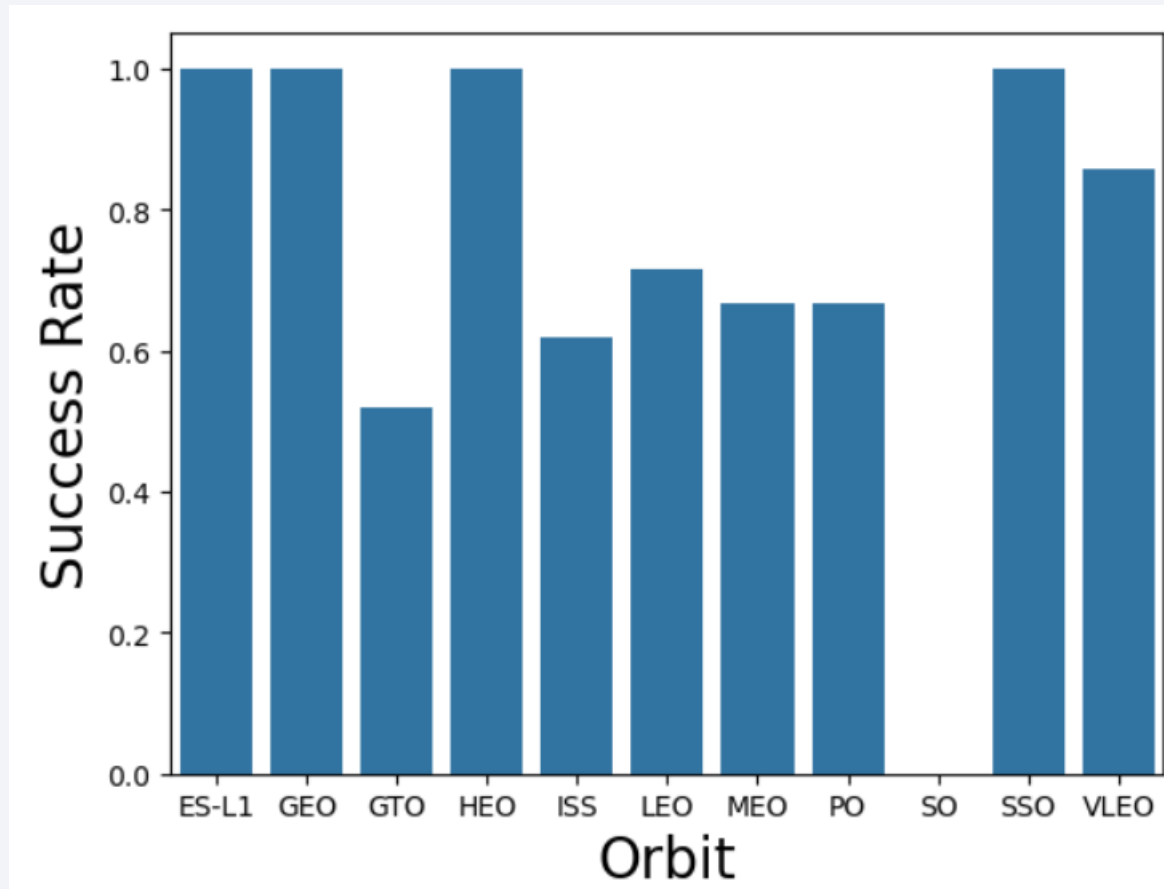
- Shows the relationship between flight number and launch site.
- In the early stages of rocket launches, launch site CCAFS SLC 40 was more frequently used, while in the mid-20s, around 25-40, launch site KSC LC 39A was more frequently used.
- The higher the flight number, the greater the probability of a successful rocket launch.

Payload vs. Launch Site



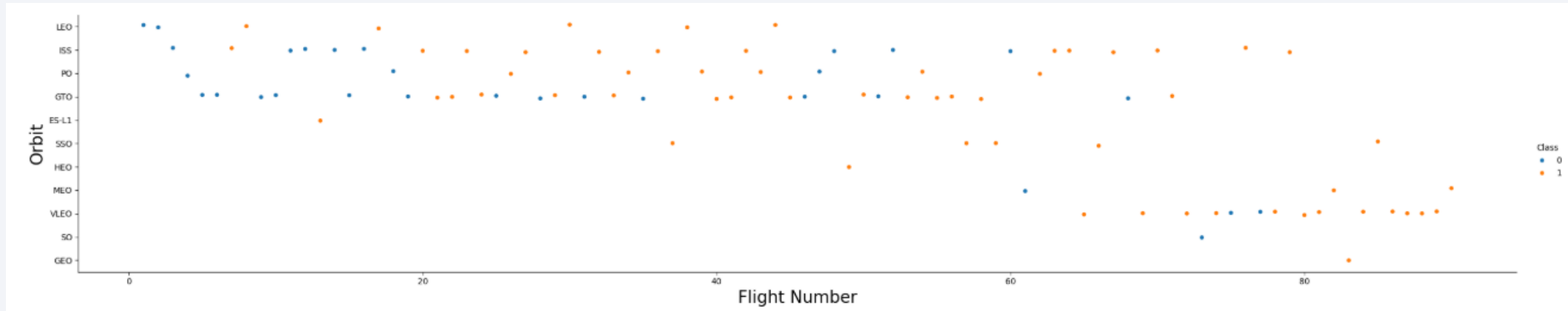
- Shows the relationship between payload mass and launch site.
- The maximum payload mass ever carried during a rocket launch at the VAFB SLC 4E launch site is less than 10,000 kg, while other launch sites can carry payloads up to 15,000 kg.
- The higher the payload, the greater the probability of a successful rocket launch.

Success Rate vs. Orbit Type

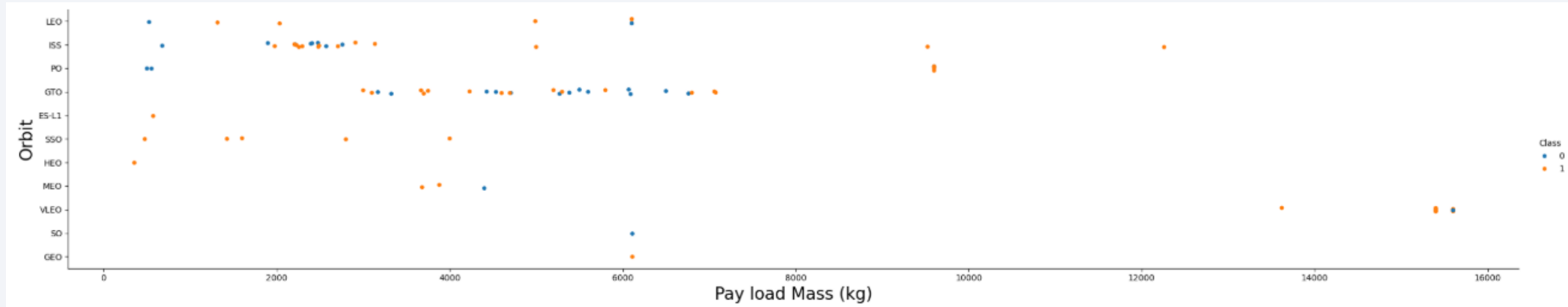


Rocket launches in ES-L1, GEO, HEO, and SSO orbits are always successful, in contrast to rocket launches in SO orbits which are never successful, while rocket launches in other orbits have a success rate of between 50 % - 85 %.

Flight Number vs. Orbit Type

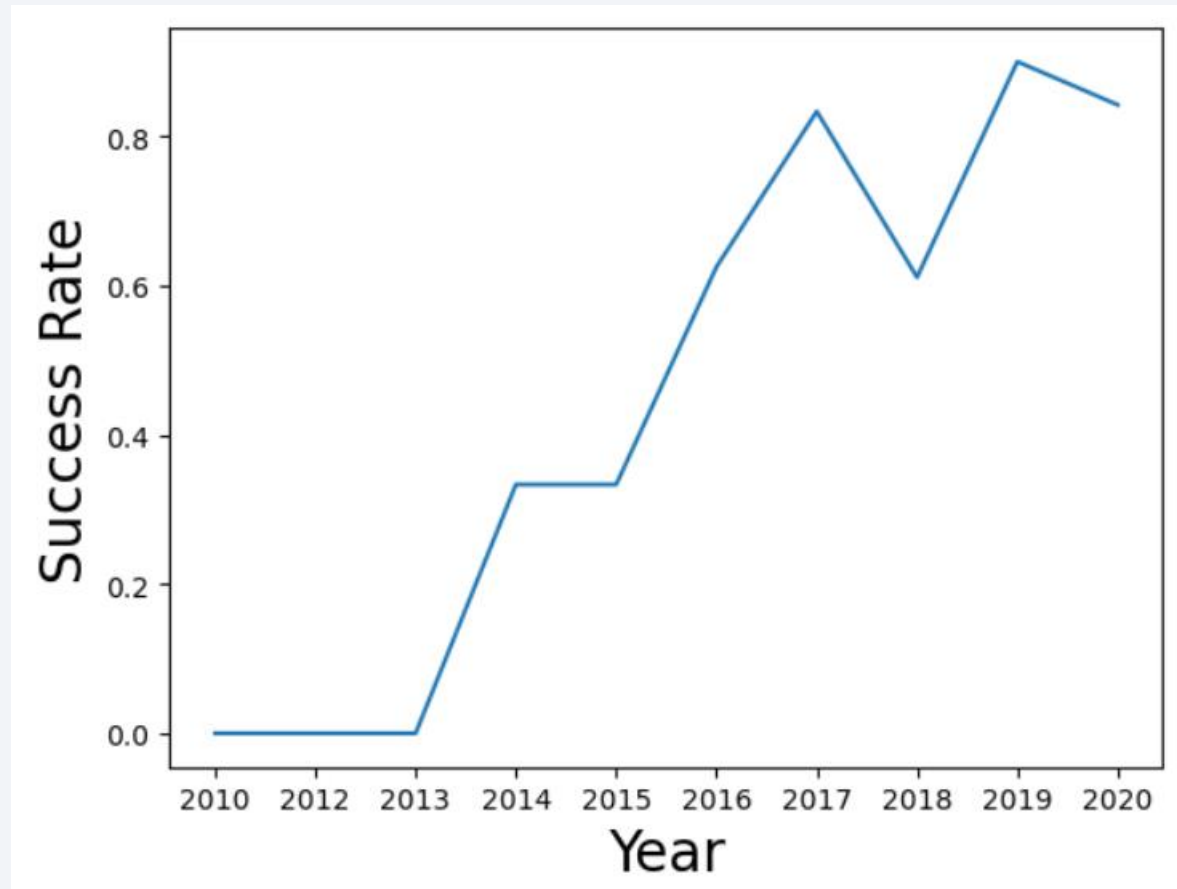


Payload vs. Orbit Type



- Shows the relationship between payload mass and orbit.
- The maximum payload mass carried during rocket launches through most orbits is usually less than 8,000 kg, while rocket launches through the ISS, PO, and VLEO orbits can carry payload masses above 8,000 kg.
- The higher the payload, the higher the probability of a successful rocket launch.

Launch Success Yearly Trend



The percentage of successful rocket launches experienced an increasing trend every year starting from 2013 - 2017 and fluctuated from 2017 - 2020.

All Launch Site Names

```
%sql select distinct launch_site from SPACEXTABLE
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Shows the name of every launch site ever used to launch a Falcon 9 rocket.

Launch Site Names Begin with 'CCA'

```
%sql select * from SPACEXTABLE where launch_site like 'CCA%' limit 5
```

Python

```
* sqlite:///my\_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Showing 5 data from Falcon 9 rocket launches launched from launch sites that have launch site names starting with 'CCA'

Total Payload Mass

```
%sql select sum(PAYLOAD_MASS_KG_) from SPACEXTABLE where Customer="NASA (CRS)"
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

```
sum(PAYLOAD_MASS_KG_)
```

```
45596
```

Displays the total payload mass carried by boosters/customers with NASA names.

Average Payload Mass by F9 v1.1

```
%sql select avg(PAYLOAD_MASS_KG_) from SPACEXTABLE where Booster_Version like "F9 v1.1%"
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

```
avg(PAYLOAD_MASS_KG_)
```

```
2534.6666666666665
```

Displays the average of all payload masses carried using the booster version with the code prefix 'F9 v1.1'

First Successful Ground Landing Date

```
%sql select min(date) from SPACEXTABLE where Landing_Outcome="Success (ground pad)"
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

```
min(date)
```

```
2015-12-22
```

Displays the date of the first successful landing of a Falcon 9 rocket on a ground pad.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql select Booster_Version from SPACEXTABLE where Landing_Outcome="Success (drone ship)" and PAYLOAD_MASS__KG_ between 4000 and 6000
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Displaying the names of booster versions that successfully landed on drone ships with a payload mass of between 4000 and 6000 kg.

Total Number of Successful and Failure Mission Outcomes

```
%sql select Mission_Outcome, count(*) from SPACEXTABLE group by Mission_Outcome
```

* [sqlite:///my_data1.db](#)

Done.

Mission_Outcome	count(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Displays the total number of successful and failed mission outcomes

Boosters Carried Maximum Payload

```
%sql select Booster_Version from SPACEXTABLE where PAYLOAD_MASS_KG_ = (select max(PAYLOAD_MASS_KG_) from SPACEXTABLE)
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

Displays the names of the booster versions that carry the maximum payload mass.

2015 Launch Records

```
%sql select Date, substr(Date,6,2) as months, Landing_Outcome, Booster_Version, Launch_Site from SPACEXTABLE where substr(Date,0,5)='2015' and Landing_Outcome="Failure (drone ship)"
```

Python

```
* sqlite:///my\_data1.db
```

Done.

Date	months	Landing_Outcome	Booster_Version	Launch_Site
2015-01-10	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Showing the date, month, landing outcome, booster version, and launch site in 2015 with landing outcome failure on drone ship

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql select Landing_Outcome, count(*) from SPACEXTABLE where Date between "2010-06-04" and "2017-03-20" group by Landing_Outcome order by count(*) desc
```

```
* sqlite:///my\_data1.db
```

```
Done.
```

Landing_Outcome	count(*)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Displays the highest to lowest ranking of the number of different landing outcomes between 2010-06-04 to 2017-03-20

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

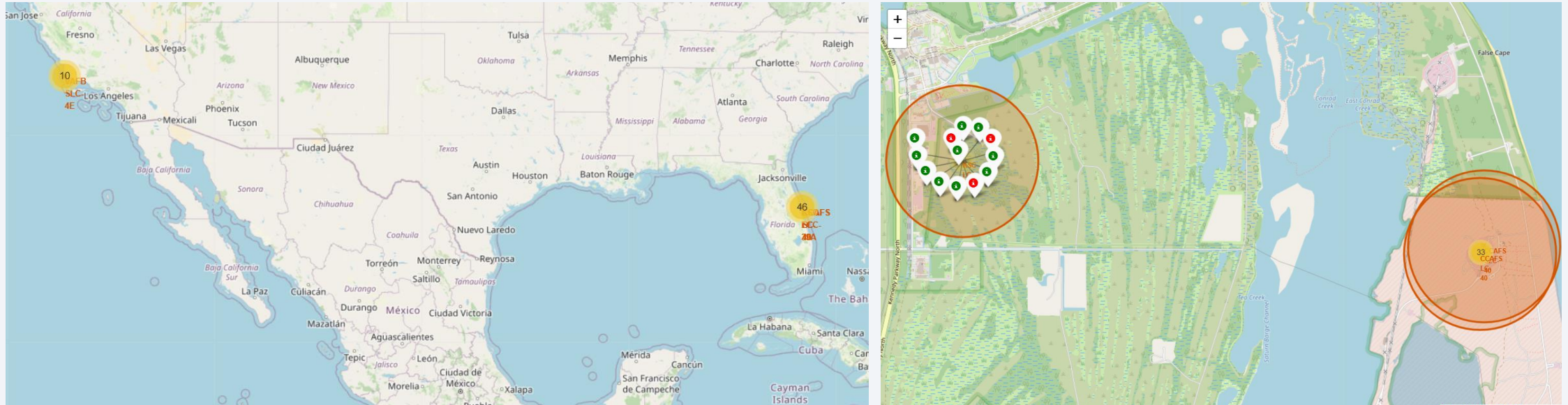
Launch Sites Proximities Analysis

Mark all launch sites on a map



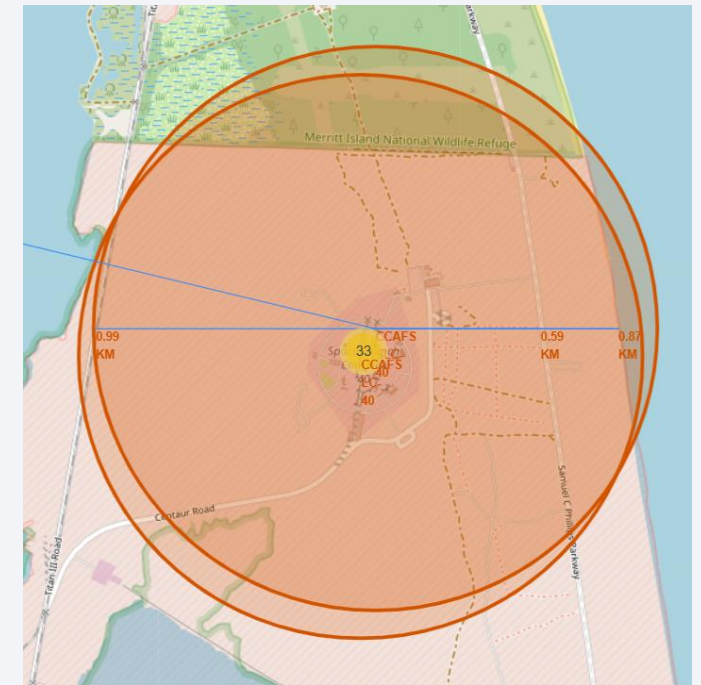
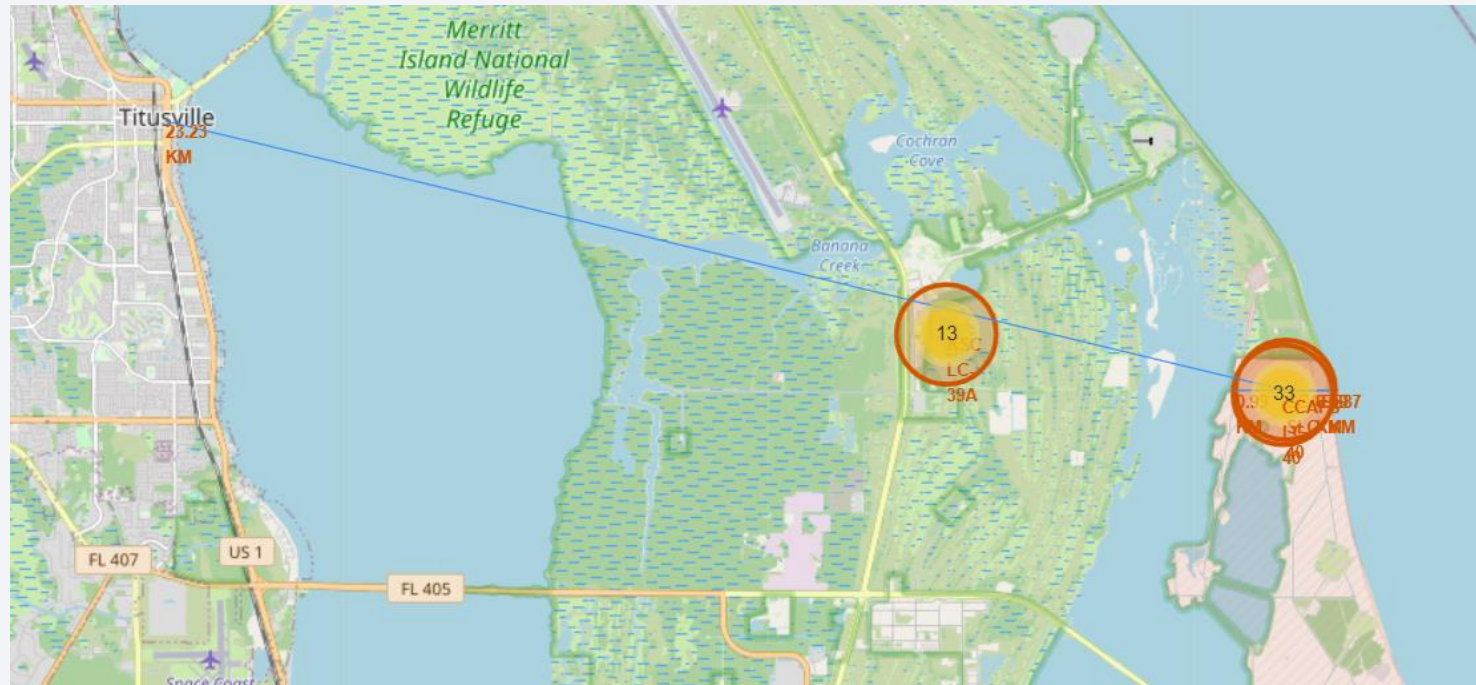
- Displays the location points of all launch sites on the map, including the name of the launch site and a 1000 m circle around it.
- All launch sites are located near the coastline.

Mark the success/failed launches for each site on the map

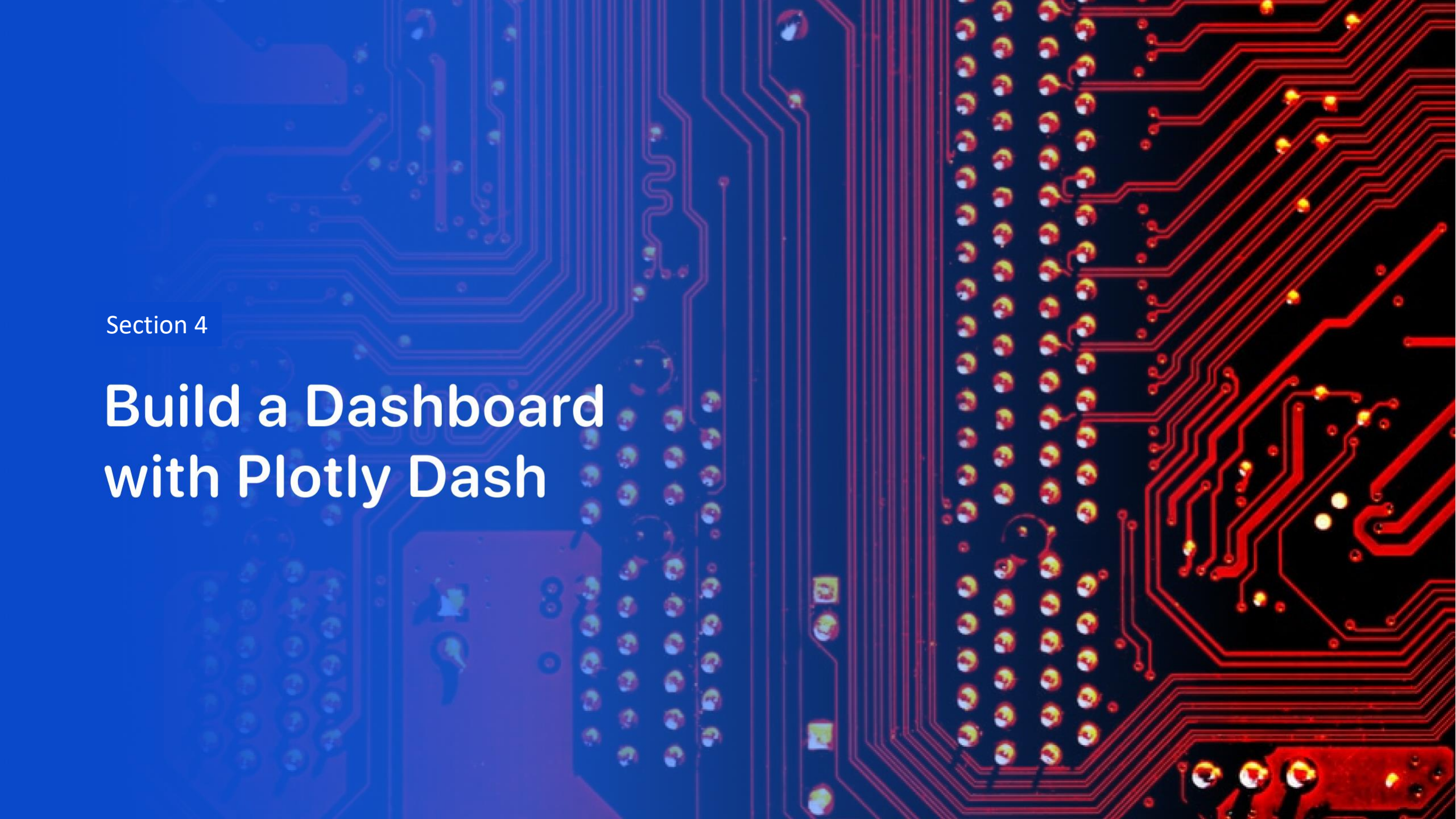


- Displays the number and status of launch outcomes from each launch site.
- Launch outcome status is represented using colors, such as green indicating a successful launch, while red indicates a failed launch.
- The KSC LC-39A launch site has a high percentage of successful launches.

Calculate the distances between a launch site to its proximities



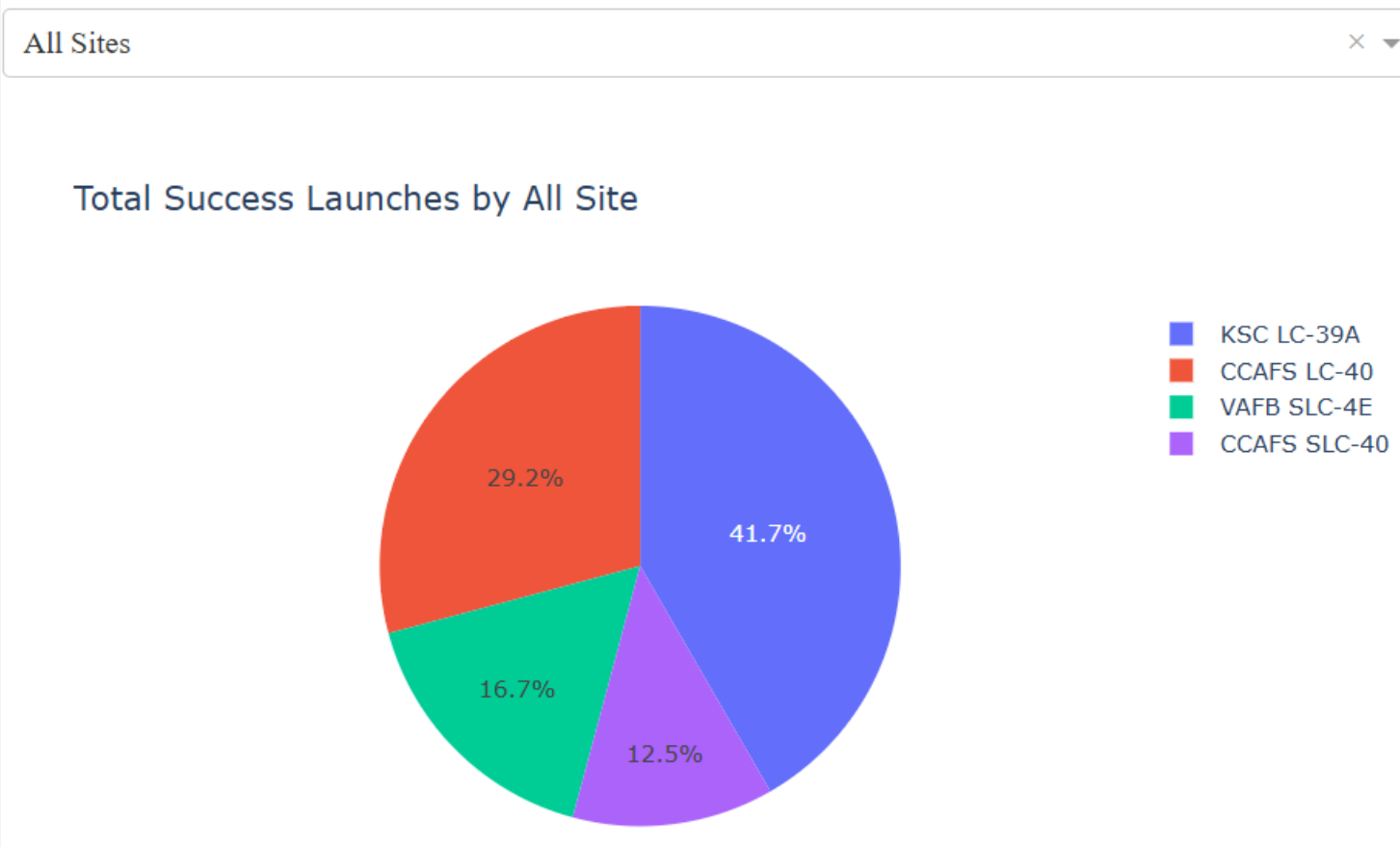
- Shows the closest distance from the CCAFS SLC-40 launch site to railways, highways, coastlines, and cities.
- The launch site is located relatively close to coastlines rather than cities because a failed rocket launch could potentially pose a hazard to the surrounding area.
- The distance to the highway (Samuel C Philips Parkway) is 0.59 km
- The closest distance to the coastline is 0.87 km
- The distance to the railway (Titan III Road) is 0.99 km
- The distance to the city (Titusville) is 23.23 km



Section 4

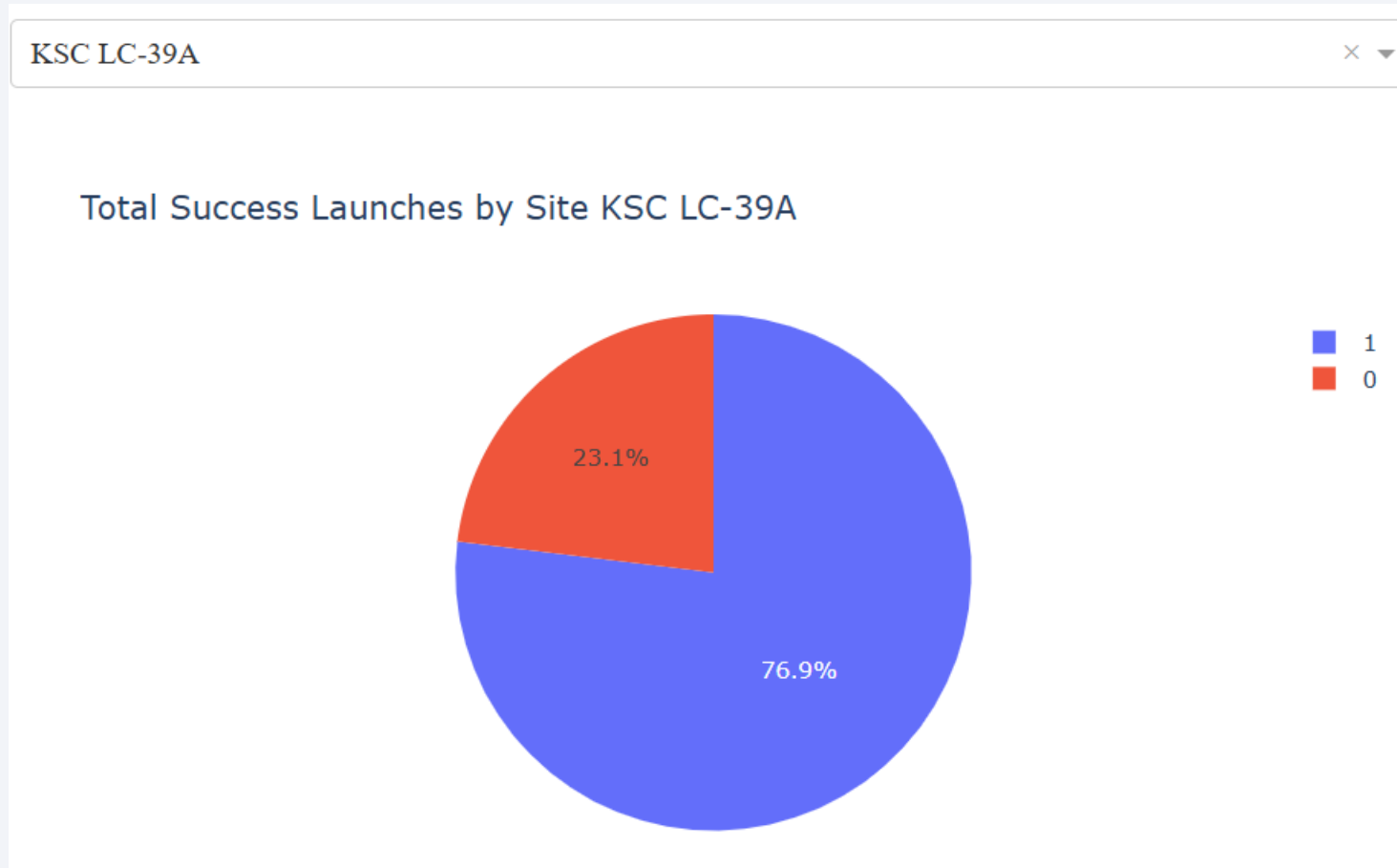
Build a Dashboard with Plotly Dash

Total Launch Success Ratio from All Site



- Showing the success rate of rocket launches across all sites.
- Site KSC LC-39A had the highest success rate at 41.7%, 12.5% higher than the second-place site. Site CCAFS SLC-40 had the lowest success rate at 12.5%.

Total launch outcome ratio of the site with the highest success rate



- Showing the launch outcome ratio from the KSC LC-39A site.
- Launches conducted at the KSC LC-39A site had a success rate of 76.9%.

Payload mass vs launch outcome

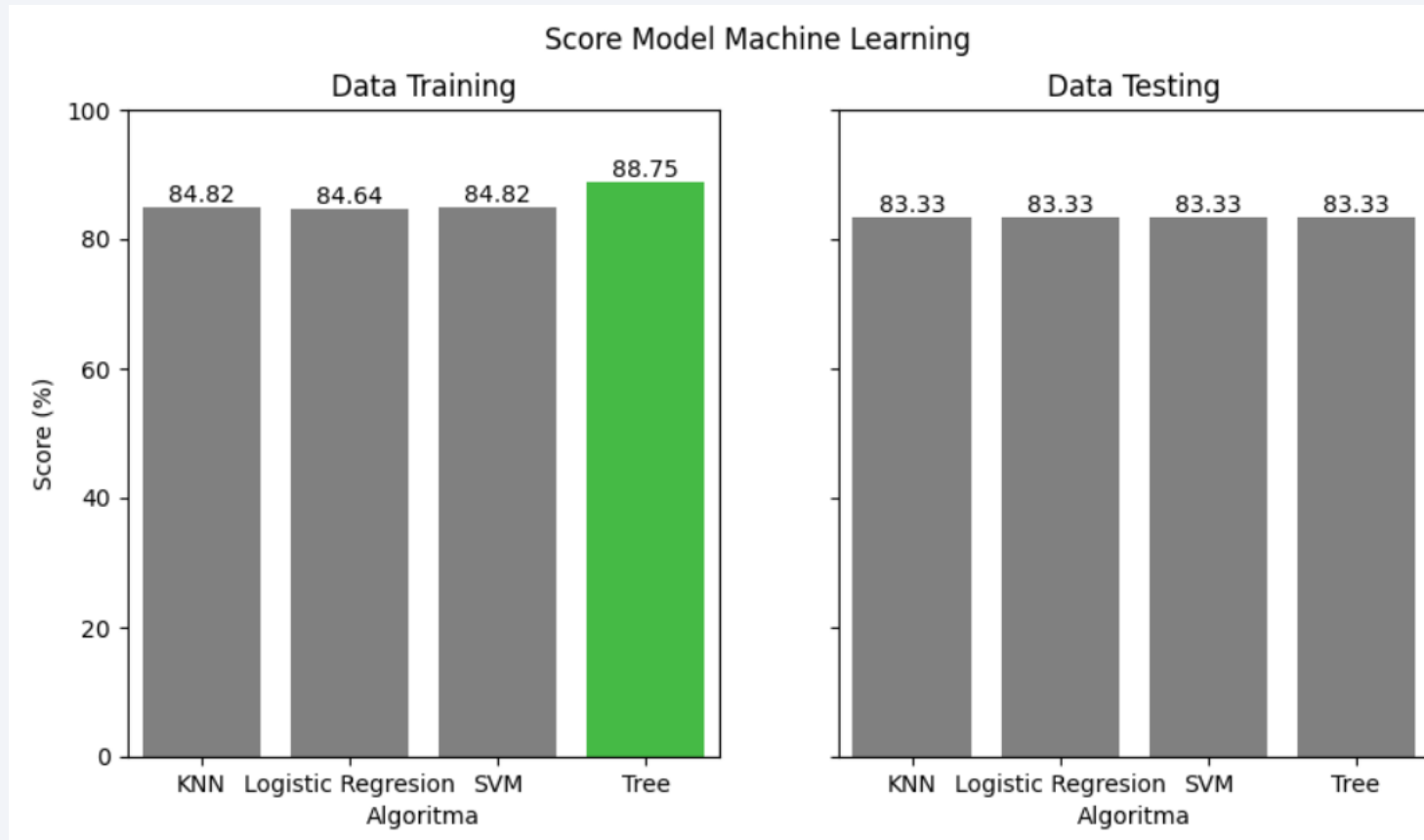


- Showing the relationship between launch outcomes and payload mass, booster version, and launch site.
- Most successful Falcon 9 rocket launches at all sites carried payload masses of approximately 2,000 kg–5,500 kg. Only the v1.0 booster version did not carry a payload mass in this range.
- At the KSC LC-39A launch site, which has a high success rate, most launches carried payload masses in the range of approximately 2,500 kg–5,500 kg with booster versions FT, B4, and B5.

Section 5

Predictive Analysis (Classification)

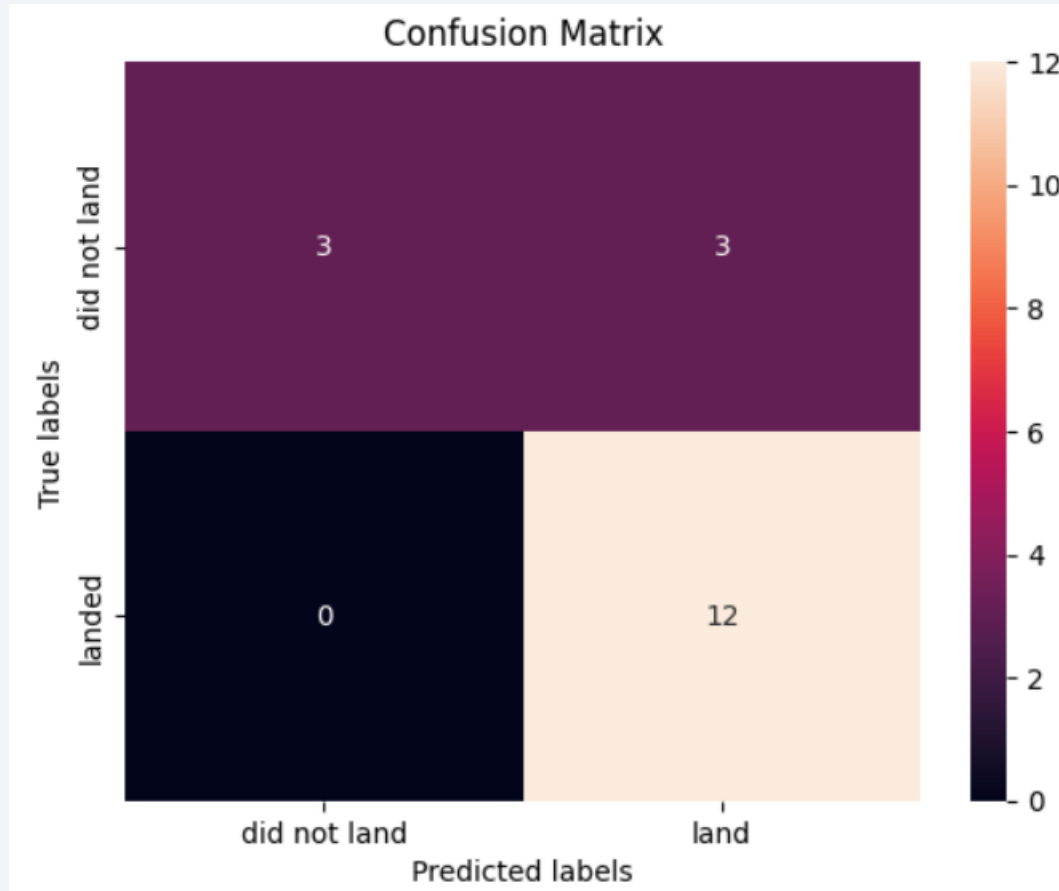
Classification Accuracy



The classification model with the highest accuracy was the Tree algorithm.

After testing all classification models using test data, they all achieved the same accuracy of 83.33%. However, when tested using training data, the accuracy varied between 84.82% and 88.75%.

Confusion Matrix

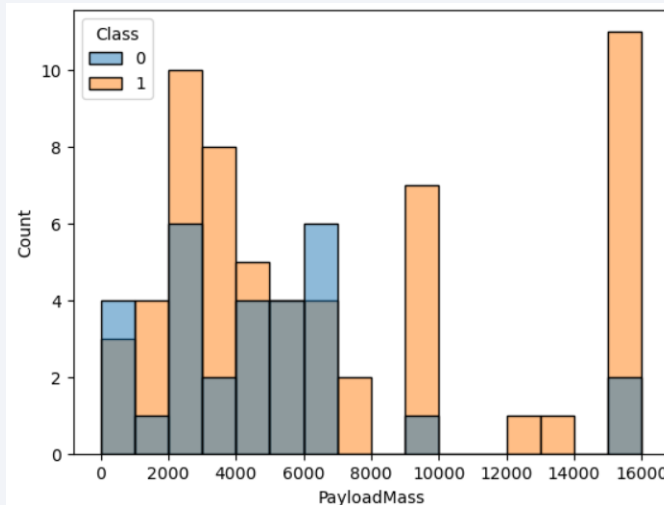
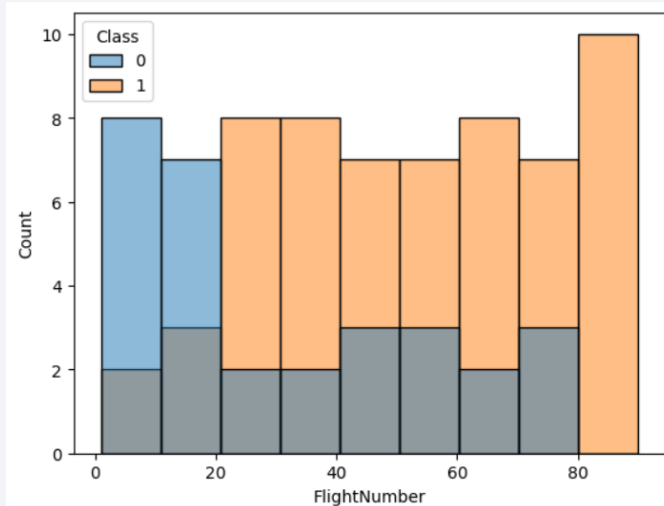


The tested data consisted of 12 launches that successfully landed, while 6 did not land, of the 18 data points tested, the classification model yielded 15 correct predictions and 3 incorrect predictions, of the 3 incorrect predictions, all launches were predicted to land but did not land.

Conclusions

- The percentage of successful rocket launches increases every year.
- The location of rocket launch sites is relatively closer to the coastline than to cities.
- The classification model with the highest accuracy is the Tree algorithm.
- Rocket launches carrying a larger payload mass have a higher probability of success.
- Rocket launches with a larger Flight Number have a higher probability of success.

Appendix



- Rocket launches with a larger Flight Number have a higher probability of success. Flight Numbers below 20 have a higher number of failed launches, while launches with Flight Numbers above 20 have a higher number of successful launches. Flight Numbers above 80 have a 100% success rate.
- Rocket launches with a larger payload mass have a higher probability of success. Rocket launches with a payload mass below 1,000 kg or between 6,000 kg and 7,000 kg have a higher number of failed launches, while rocket launches with a payload mass between 1,000 kg and 5,000 kg or above 7,000 kg have a higher number of successful launches.

Thank you!

